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Loss of outer membrane protein A (OmpA) impairs the survival of *Salmonella* Typhimurium by inducing membrane damage in the presence of ceftazidime and meropenem

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Objectives: *Salmonella enterica* serovar Typhimurium is one of the significant non-typhoidal *Salmonella* serovars that causes gastroenteritis. The rapid development of antimicrobial resistance necessitates studying new antimicrobials and their therapeutic targets in this pathogen. Our study aimed to investigate the role of four prominent outer membrane porins of *S. Typhimurium*, namely OmpA, OmpC, OmpD and OmpF, in developing resistance against ceftazidime and meropenem.

Methods: The antibiotic-mediated inhibition of bacterial growth was determined by measuring the absorbance and the resazurin assay. DiBAC₄ (Bis-(1,3-Dibutylbarbituric Acid)Trimethine Oxonol), 2,7-dichlorodihydrofluorescein diacetate (DCFDA) and propidium iodide were used to determine the outer membrane depolarization, reactive oxygen species (ROS) generation and subsequent killing of *Salmonella*. The expression of oxidative stress-response and efflux pump genes was quantified by quantitative RT-qPCR. HPLC was done to determine the amount of antibiotics that entered the bacteria. The damage to the bacterial outer membrane was studied by confocal and atomic force microscopy. The *in vivo* efficacy of ceftazidime and meropenem were tested in the C57BL/6 mouse model.

Results: Deleting *ompA* reduced the survival of *Salmonella* in the presence of ceftazidime and meropenem. Massive outer membrane depolarization and reduced expression of oxidative stress-response genes in *S. Typhimurium* $\Delta ompA$ hampered its growth in the presence of antibiotics. The enhanced uptake of antibiotics and decreased expression of efflux pump genes in *S. Typhimurium* $\Delta ompA$ resulted in damage to the bacterial outer membrane. The clearance of the *S. Typhimurium* $\Delta ompA$ from C57BL/6 mice with ceftazidime treatment proved the role of OmpA in rendering protection against β -lactam antibiotics.

Conclusions: OmpA protects *S. Typhimurium* from two broad-spectrum β -lactam antibiotics, ceftazidime and meropenem, by maintaining the stability of the outer membrane.

Introduction

Salmonella enterica serovar Typhimurium, the most commonly reported non-typhoidal serovar of the Gram-negative pathogen *Salmonella enterica*, causes self-limiting gastroenteritis in humans and typhoid fever-like symptoms in animals.^{1–3} Every year, 1.3 billion cases of *Salmonella*-related gastroenteritis are reported globally, with approximately 3 million deaths.⁴ *S. Typhimurium* ST19 and ST34 are the primary reasons for global gastroenteritis.⁵ However, bloodstream infection caused by non-typhoidal *Salmonella* serovars is also a serious health hazard in

sub-Saharan Africa (SSA).^{6–9} Over the past 30 years, the rapid emergence and subsequent global spread of MDR *Salmonella* Typhimurium, such as *S. Typhimurium* DT104, has posed a severe threat to public health.¹⁰

The outer membrane of Gram-negative bacteria is densely populated with porins, outer membrane-bound β -barrel proteins, which help in transporting salts, sugars, nutrients, peptides, amino acids, vitamins etc.^{11,12} Bacteria use porins to change their outer membrane permeability to develop resistance against antibiotics, antimicrobial peptides etc.^{13–15} Many Gram-negative pathogens require outer membrane protein A

Table 1. Strains used in this study

Strains/plasmids	Characteristics	Source/references
<i>S. Typhimurium</i> ATCC strain 14028	WT	Gifted by Prof. M. Hensel
<i>S. Typhimurium</i> $\Delta ompA$	Kan ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i>	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompC$	Chl ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompD$	Kan ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompF$	Chl ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L1-1	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L1-2	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L2-1	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L2-2	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L3-1	Kan ^R , Amp ^R	Laboratory stock
<i>S. Typhimurium</i> $\Delta ompA$:pQE60- <i>ompA</i> L4-1	Kan ^R , Amp ^R	Laboratory stock

(OmpA), one of the most abundant outer membrane porins in the Enterobacteriaceae family, to develop antimicrobial resistance.^{15–18} Apart from OmpA, the outer membrane of *Salmonella* has other important porins such as OmpC, OmpD and OmpF.¹⁹ The loss of OmpC protects *Salmonella* from cephalosporins.²⁰ On the contrary, clinical isolates of *Salmonella* Typhi expressing OmpC are resistant to chloramphenicol.²¹ OmpF deficiency is associated with imipenem resistance in *Salmonella enterica* serovar Wien and chloramphenicol resistance in *Salmonella* Typhi.^{21,22}

In the current study, we investigated the specific contribution of OmpA, OmpC, OmpD and OmpF of *Salmonella* Typhimurium in developing resistance against two broad-spectrum β -lactam antibiotics, ceftazidime and meropenem.^{23–28} Our data revealed that deleting OmpA from *Salmonella* increased the outer membrane permeability and facilitated the entry of antibiotics, which further disrupted its cellular morphology. Reduced expression of *marA* (transcriptional activator), *acrF* (efflux pump) and several oxidative stress response genes (*katG*, *katE*, *sodA* and *sodB*) in *S. Typhimurium* $\Delta ompA$ made it vulnerable to antibiotic treatment. To the best of our knowledge, this is the first study demonstrating the protective role of *S. Typhimurium* OmpA against ceftazidime and meropenem.

Materials and methods

Bacterial strains, media, and culture conditions

All the bacterial strains used in this study were revived from glycerol stock (stored at -80°C) and plated on LB agar with or without antibiotics (Table 1). A single bacterial colony from the LB agar plates was inoculated into LB broth. The stationary phase culture was further subcultured at a 1:100 ratio in a fresh LB tube and incubated till the mid-log phase, followed by normalization of OD₆₀₀ to 0.1 ($\sim 10^6$ cfu) before performing all experiments.

Determination of the MIC of ceftazidime and meropenem

The log phase cultures of different bacterial strains were incubated in the wells of a 96-well plate containing Cation-adjusted Muller-Hinton Broth (CA MHB) in the presence of either ceftazidime (concentrations 128, 64, 32, 16, 8, 4, 2, 1, 0.5, 0.25 and 0 mg/L) or meropenem (concentrations

8, 4, 2, 1, 0.5, 0.25, 0.125, 0.06, 0.03, 0.01 and 0 mg/L) in a shaking incubator at 37°C temperature at 170 rpm for 18 to 24 h. At the end of the incubation period, the MIC was determined by OD₆₀₀ measurement in a TECAN microplate reader.

Determination of bacterial viability by resazurin assay

To the antibiotic-treated bacterial cells, 20 μL of resazurin solution from 0.2 mg/mL stock was added and incubated for 2 h in the dark. The fluorescence intensity of resazurin (excitation 540 nm and emission 590 nm) was measured with a TECAN microplate reader, and the percent viability was calculated.

Determination of reactive oxygen species (ROS) generation, outer membrane depolarization, and killing of *Salmonella* by flow cytometry

The antibiotic-treated bacterial cells were incubated with 2,7-dichlorodihydrofluorescein diacetate (DCFDA) (10 μM), Bis-(1,3-Dibutylbarbituric Acid)Trimethine Oxonol (DiBAC₄) (1 $\mu\text{g}/\text{mL}$) and propidium iodide (1 $\mu\text{g}/\text{mL}$) for 15 min to measure the antibiotic-mediated ROS generation, outer membrane depolarization and killing of *Salmonella*. The cells were immediately subjected to analysis by flow cytometry (BD FACSVerser by BD Biosciences, USA). The results were analysed by BD FACSuite software.

RNA isolation and quantitative RT-PCR (RT-qPCR)

Overnight grown bacterial cultures were centrifuged at 13000 rpm for 20 min, and the pellets were resuspended in TRIzol solution. The lysed supernatants were subjected to chloroform and isopropanol extraction subsequently. The RNA pellet was washed with 70% ethanol and resuspended in DEPC water. After an hour of digesting the RNA with RNase-free DNase, the cDNA was prepared using the PrimeScript™ cDNA synthesis kit following the manufacturer's protocol. The cDNA was used to quantify the transcript-level expression of our target genes using appropriate oligos (Table 2), SYBR/TB green qRT-PCR kit, and RT-qPCR detection system.

Measuring the entry of meropenem by HPLC

HPLC from Agilent Technologies (1120 Compact LC) with a C18 column as a stationary phase was used to quantify the entry of meropenem in the bacteria. A mixture of 0.1% aqueous acetic acid (solution A) and methanol

Table 2. The list of oligonucleotides used in qRT-PCR

Genes	Primer	Sequence (5'–3')
<i>sodA</i>	Forward	GCGTGTCCACACATCAAC
<i>sodA</i>	Reverse	GCTATCGCGGCATCTTTGG
<i>sodB</i>	Forward	CCTGCCGGTTGAAGAACTGA
<i>sodB</i>	Reverse	ACCGAAGTCACGCTCGATAG
<i>katG</i>	Forward	GACTCACCGACACCCTGAAG
<i>katG</i>	Reverse	CACGGTCTCTTCGTCGTTCA
<i>katE</i>	Forward	TGAGCCAGCGTGACATCAAT
<i>katE</i>	Reverse	CAGCCCTCTTTCGGTGAAT
<i>acrB</i>	Forward	GATTACAGGCGCATTTGGTGG
<i>acrB</i>	Reverse	TCGGTGTAGTGATGCGTGCT
<i>acrF</i>	Forward	GGGAGGACTCGGTCACGATG
<i>acrF</i>	Reverse	CCGCGGAGATGGTTTCGTTG
<i>marA</i>	Forward	GGCACCTGCAACGGATGTTT
<i>marA</i>	Reverse	AGCGGCAGCATATACCGTGA
16S rRNA	Forward	GAGCGCAACCTTATCCTTTG
16S rRNA	Reverse	CACTTTATGAGGTCCGCTTGCT

Table 3. The gradient programme of the mobile phase used for HPLC

Time (min)	Solution A (0.1% aqueous acetic acid)	Solution B (methanol)
0	95%	5%
2	95%	5%
18	75%	25%
20	75%	25%
25	50%	50%
28	50%	50%
29	95%	5%
35	95%	5%

(solution B) was used as a mobile phase with a gradient programme (Table 3). The flow rate of the mobile phase was kept at 1 mL/min for chromatographic separation. The detection wavelength for meropenem was fixed at 300 nm. Ceftazidime was quantified by a protocol as demonstrated earlier, using HPLC equipment from SHIMADZU.²⁹ An isocratic solution of acetonitrile and acetic acid (75:25) with a flow rate of 0.7 mL/min was used as the mobile phase, and the detection of ceftazidime was done at 256 nm. The exponentially growing bacterial cultures were treated with high doses of antibiotics (meropenem ~ 100–150 mg/L, ceftazidime 300 mg/L) for 1 h. Post-treatment, the cells were centrifuged at 5000 rpm for 20 min. Twenty microlitres of the filter-sterilized culture supernatant was used to quantify the amount of remaining antibiotics. The results were analysed by EZChrom Elite or Shimadzu software, and the AUC was measured to estimate the concentration of available antibiotics in the media.

Assessing the damage to the bacterial outer membrane by confocal microscopy

The antibiotic-treated or untreated bacterial cells were fixed with 3.5% paraformaldehyde (PFA). To stain the outer membrane of the bacteria, the fixed bacterial cells were incubated with FM 4-64 dye (excitation/emission: 514/640 nm) (0.01 μ g/mL) for 30 min. The DNA of the bacterial cells was visualized by DAPI staining (0.01 μ g/mL). The images were

acquired and analysed by confocal laser scanning microscopy 880 (ZEISS) and ZEN black software.

Assessing the bacterial morphology by atomic force microscopy (AFM)

The bacterial cells were treated with a sublethal concentration of meropenem (0.03 mg/L) for 16 to 18 h and fixed with 3.5% PFA. One hundred microlitres of diluted sample was dropcasted on sterile glass coverslips. The air-dried sample was taken to an NX10 AFM for image acquisition. The length and breadth of the bacteria were measured using XEI software.

Determining the fraction of antibiotic-tolerant persisters

The log phase culture of *S. Typhimurium* (WT) and Δ ompA were treated with 50X and 25X MIC of ceftazidime and meropenem. The persister population was obtained by plating the aliquots collected at indicated timepoints post-treatments. Similarly, 24h-old biofilm cultures were exposed to similar concentrations of antibiotics, and persister populations were enumerated by plating. The cfu obtained by plating was normalised with pre-inoculum to calculate the percent viability.

Animal experiments

C57BL/6 mice were infected with 10^6 cfu of *S. Typhimurium* (WT) and Δ ompA. On the second and fourth day post-infection, the infected mice were intraperitoneally administered with ceftazidime and meropenem (5 mg/kg of body weight) separately. Antibiotic-untreated mice were kept as controls. On the fifth day post-infection, the mice were sacrificed, and the liver and spleen from infected mice were collected. The organs were homogenized with sterile glass beads, and the lysates were plated on *Salmonella-Shigella* agar plates to enumerate the bacterial load. The cfu obtained after plating was normalized with the weight of the individual organs, and the log values of the normalized cfu were plotted.

Statistical analysis

Each experiment has been independently repeated with at least two biological replicates. The graphs and the statistics were generated by GraphPad Prism 8.4.3 software. For multiple comparisons, two-way ANOVA was used. To quantify the HPLC data, the estimation of percent viability for the OmpA loop mutants and the calculation of the persister population unpaired Student's *t*-test were used. In all the cases, *P* values below 0.05 were considered significant.

Ethics

All the animal experiments performed in this study were approved by the Institutional Animal Ethics Committee (IAEC) at the Indian Institute of Science, Bangalore, India (Registration No: 48/1999/CPCSEA), and the guidelines provided by the CPCSEA were followed. The Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA) was formed under Chapter 4, Section 15(1) of the Prevention of Cruelty to Animals Act 1960. The ethical clearance number for this study is CAF/Ethics/670/2019.

Results

Deleting OmpA from *S. Typhimurium* enhanced the susceptibility of the bacteria to ceftazidime and meropenem

To investigate the role of outer membrane porins in antimicrobial resistance of *Salmonella*, *ompA*, *ompC*, *ompD* and *ompF* mutant

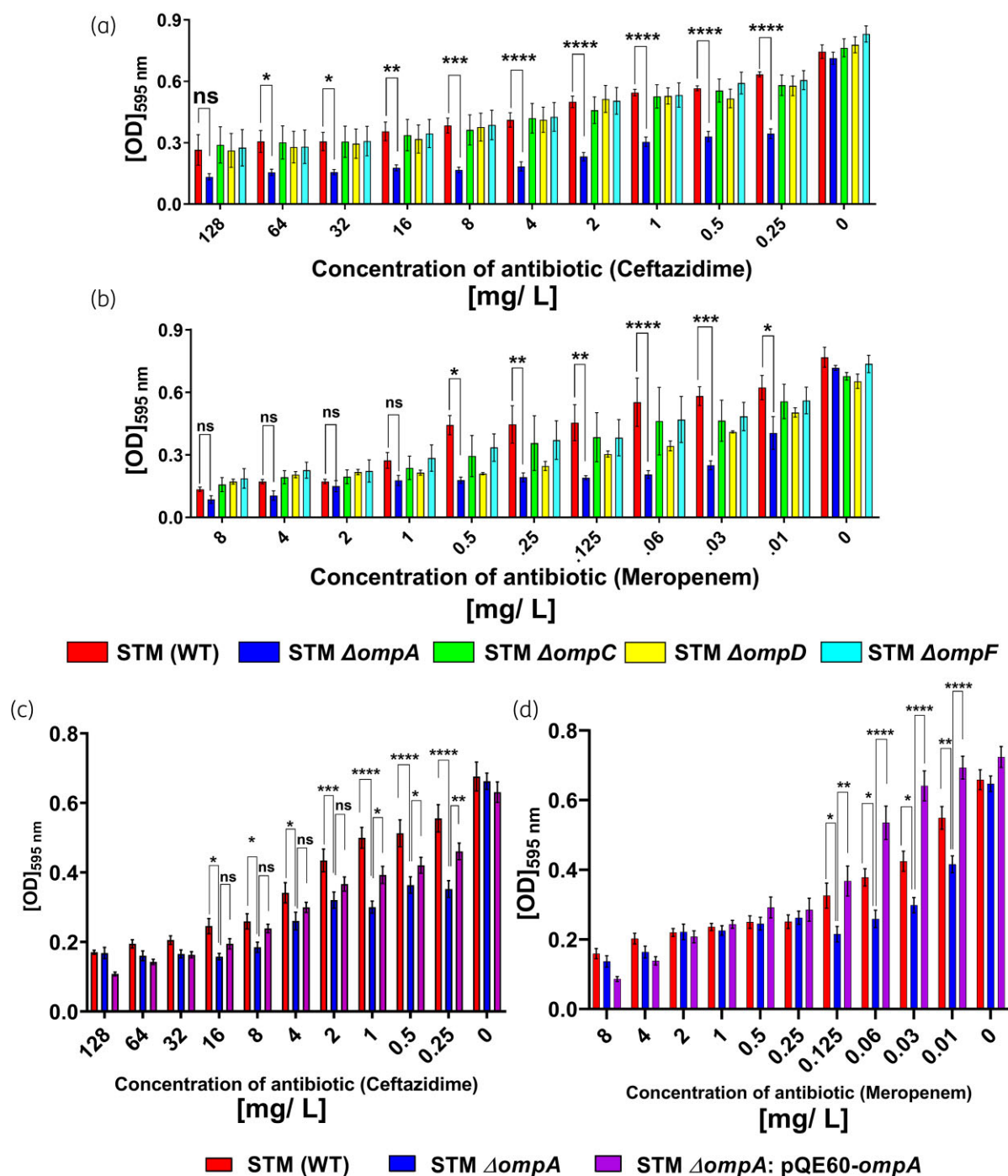


Figure 1. Deleting OmpA from *S. Typhimurium* enhanced the susceptibility of the bacteria to ceftazidime and meropenem. Determination of MICs of (a) ceftazidime ($N=7$) and (b) meropenem ($N=4$) for *S. Typhimurium* (WT), $\Delta ompA$, $\Delta ompC$, $\Delta ompD$ and $\Delta ompF$ growing in CA MHB. Studying the growth inhibition of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA$:pQE60-*ompA* growing in CA MHB in the presence of varying concentrations of (c) ceftazidime ($N=8$) and (d) meropenem ($N=5$). Data are presented as mean $\pm$ SEM. * $P<0.05$; ** $P<0.005$; *** $P<0.0005$; **** $P<0.0001$; ns, non-significant (two-way ANOVA). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

strains were incubated with ceftazidime and meropenem for 18 to 24 h. The enhanced susceptibility and reduced MICs of ceftazidime and meropenem for *S. Typhimurium* $\Delta ompA$ suggested a protective role of OmpA against both antibiotics [Figure 1(a and b), Tables 4 and 5]. Surprisingly, deleting OmpC, OmpD and OmpF did not show any significant growth inhibition of

Table 4. Concentrations of ceftazidime used in Mueller–Hinton broth for testing its susceptibility against *S. Typhimurium* (WT), $\Delta ompA$, $\Delta ompC$, $\Delta ompD$ and $\Delta ompF$

Concentration of ceftazidime (mg/L)	STM (WT)		STM $\Delta ompA$		STM $\Delta ompC$		STM $\Delta ompD$		STM $\Delta ompF$	
	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)
0	0.7447	100	0.7128	100	0.7634	100	0.7782	100	0.8310	100
0.25	0.6335	74.69	0.3439	48.91	0.5809	75.24	0.5780	99.69	0.6058	101.4
0.5	0.5656	80.90	0.3305	49.08	0.5545	69.54	0.5159	95.42	0.5915	95.73
1	0.5450	70.62	0.3042	44.00	0.5260	79.20	0.5284	85.98	0.5329	87.99
2	0.4995	72.67	0.2319	41.37	0.4585	75.32	0.5137	90.39	0.5043	93.79
4	0.4116	85.81	0.1841	48.10	0.4194	85.36	0.4125	94.56	0.4262	96.01
8	0.3841	96.38	0.1667	34.26	0.3627	82.14	0.3763	95.67	0.3864	107.0
16	0.3548	99.06	0.1776	18.55	0.3371	78.35	0.3179	93.25	0.3449	98.02
32	0.3060	98.35	0.1565	11.42	0.3048	83.37	0.2955	90.56	0.3078	100.6
64	0.3059	99.38	0.1553	12.12	0.3011	85.82	0.2786	92.77	0.2802	99.25
128	0.2650	98.01	0.1334	9.912	0.2889	82.31	0.2625	94.56	0.2754	103.5

STM, *S. Typhimurium*.

Salmonella when incubated with antibiotics [Figure 1(a and b), Tables 4 and 5]. In addition to this, the results from the resazurin assay suggested a significant reduction in the percent viability of *S. Typhimurium* $\Delta ompA$ relative to *S. Typhimurium* (WT) at 0.5 mg/L [Figure S1(a and b), available as Supplementary data at JAC Online, and Table 4] and 0.125 mg/L [Figure S2(a and b), and Table 5] concentration of ceftazidime and meropenem. The MICs of ceftazidime and meropenem for WT *Salmonella* spp. are 1 and 2 mg/L, respectively.³⁰ Our data suggested that deleting OmpA from *S. Typhimurium* reduced the MIC of ceftazidime to 0.5 mg/L and of meropenem to 0.125 mg/L (Tables 4 and 5). On the contrary, the *ompC*, *ompD* and *ompF* knockout strains exhibited uninterrupted growth in increasing concentrations of both

antibiotics, endorsing the dispensability of these porins in developing antimicrobial resistance against β -lactam antibiotics. The partial recovery of the bacterial growth of the *ompA*-complemented strain in sublethal concentrations of ceftazidime and meropenem strongly supported our conclusion [Figure 1(c and d)].

Increased outer membrane depolarization and reduced expression of oxidative stress response genes facilitated the killing of *S. Typhimurium* $\Delta ompA$ in the presence of antibiotics

Antimicrobial peptides kill bacterial pathogens by causing depolarization of the outer membrane.^{31–35} We also hypothesized that the

Table 5. Concentrations of meropenem used in Mueller–Hinton broth for testing its susceptibility against *S. Typhimurium* (WT), $\Delta ompA$, $\Delta ompC$, $\Delta ompD$ and $\Delta ompF$

Concentration of meropenem (mg/L)	STM (WT)		STM $\Delta ompA$		STM $\Delta ompC$		STM $\Delta ompD$		STM $\Delta ompF$	
	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)	OD ₅₉₅ (mean)	Viability (%) (mean)
0	0.7679	100	0.7165	100	0.6767	100	0.6531	100	0.7359	100
0.01	0.6225	99.16	0.4045	100.6	0.5563	100.1	0.5032	102.5	0.5605	93.49
0.03	0.5814	89.42	0.2499	58.95	0.4636	77.12	0.4098	89.38	0.4838	93.90
0.06	0.5520	80.52	0.2054	53.60	0.4618	75.79	0.3427	86.76	0.4693	89.12
0.125	0.4542	88.40	0.1900	37.54	0.3843	75.73	0.3037	79.81	0.3819	93.27
0.25	0.4457	84.98	0.1920	29.44	0.3558	78.20	0.2465	78.45	0.3704	83.23
0.5	0.4427	78.02	0.1780	17.97	0.2942	75.12	0.2098	86.31	0.3345	92.88
1	0.2730	86.24	0.1770	14.44	0.2374	76.77	0.2143	78.76	0.2844	86.24
2	0.1721	83.78	0.1495	11.77	0.1946	84.80	0.2169	83.44	0.2224	92.68
4	0.1711	81.66	0.1046	7.261	0.1923	81.85	0.2046	83.57	0.2263	95.97
8	0.1348	83.53	0.08570	6.268	0.1568	72.22	0.1716	86.44	0.1866	97.16

STM, *S. Typhimurium*.

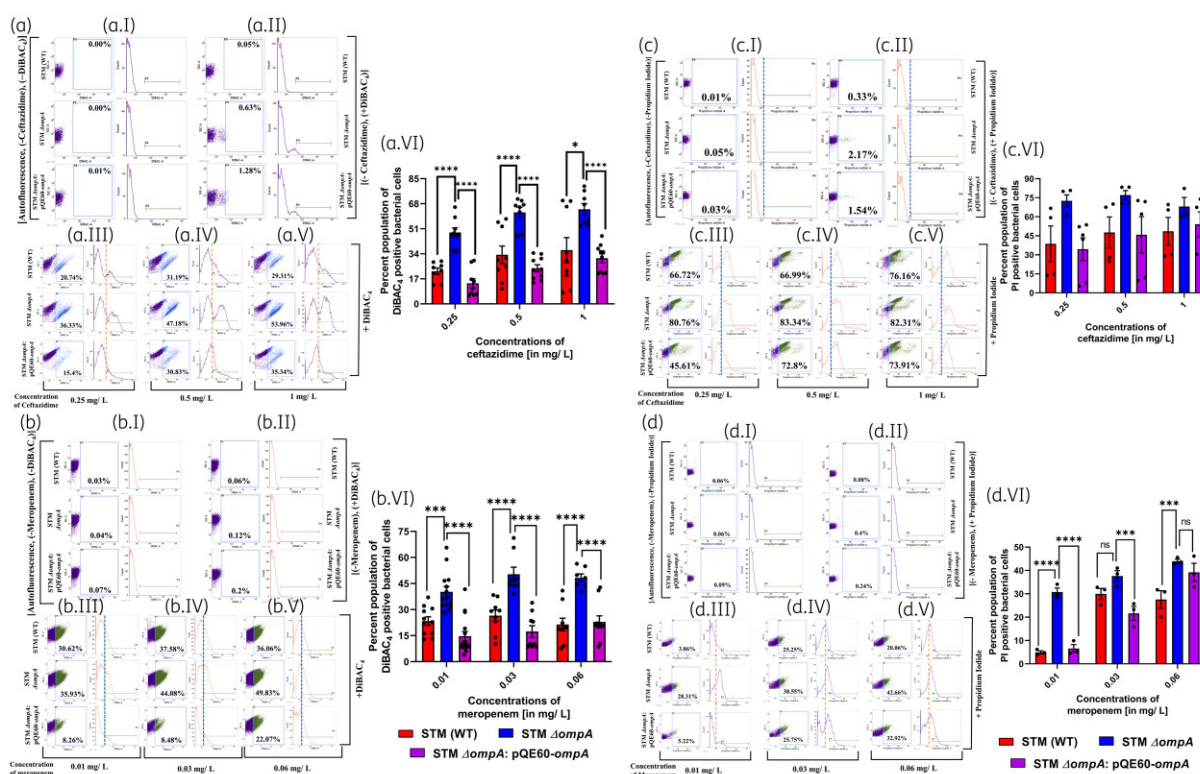


Figure 2. The increased outer membrane depolarization of *S. Typhimurium* $\Delta ompA$ induced the killing of the bacteria in the presence of antibiotics. Measuring the outer membrane depolarization of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA:pQE60-ompA$, growing in Mueller–Hinton broth in the presence of increasing concentrations of (a) ceftazidime [(a.I) autofluorescence; (a.II) non-antibiotic control; (a.III) 0.25 mg/L; (a.IV) 0.5 mg/L; (a.V) 1 mg/L; and (a.VI) cumulative trend; $n=2$, $N=5$] and (b) meropenem [(b.I) autofluorescence; (b.II) non-antibiotic control; (b.III) 0.01 mg/L; (b.IV) 0.03 mg/L and (b.V) 0.06 mg/L; (b.VI) cumulative trend, $n=2$, $N=7$] by DiBAC₄ (1 μ M) staining by flow cytometry. Quantifying the death of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA:pQE60-ompA$, growing in Mueller–Hinton broth in the presence of increasing concentrations of (c) ceftazidime [(c.I) autofluorescence; (c.II) non-antibiotic control; (c.III) 0.25; (c.IV) 0.5; (c.V) 1 mg/L; and (c.VI) cumulative trend, $n=2$, $N=3$] and (d) meropenem [(d.I) autofluorescence; (d.II) non-antibiotic control; (d.III) 0.01 mg/L; (d.IV) 0.03 mg/L; (d.V) 0.06 mg/L; and (d.VI) cumulative trend, $N=3$] by propidium iodide (1 μ M) staining by flow cytometry. The representative image corresponds to one single experiment from the independently done experiments. The dot plot (SSC-A versus DCFDA-A) and the histogram (count versus DCFDA-A) have been obtained from BD FACSuite software. Data are represented as mean $\pm$ SEM. * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; **** $P < 0.0001$; ns, non-significant (two-way ANOVA). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

enhanced antibiotic-dependent killing of *S. Typhimurium* $\Delta ompA$ happens because of the higher outer membrane depolarization. The extent of depolarization was measured by quantifying the DiBAC₄-positive population of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA:pQE60-ompA$ under antibiotic treatment by flow cytometry [Figure 2(a and b)]. Our results suggested that deleting *OmpA* did not depolarize the outer membrane of *Salmonella* without antibiotics [Figure 2(a.II and b.II)]. But when *S. Typhimurium* $\Delta ompA$ was incubated with ceftazidime [Figure 2(a.III, a.IV, a.V and a.VI)], massive outer membrane depolarization was observed [Figure 2(a.III) 36.33%; 2(a.IV) 47.18%; 2(a.V) 53.96%; and 2(a.VI) cumulative trend] compared with the WT [Figure 2(a.III) 20.74%; 2(a.IV) 31.19%; 2(a.V) 29.31%; and 2(a.VI) cumulative trend] and the complemented strains [Figure 2(a.III) 15.4%; 2(a.IV) 30.83%; 2(a.V) 35.34%; and 2(a.VI) cumulative trend]. Likewise, increased outer membrane depolarization for *S. Typhimurium* $\Delta ompA$ [Figure 2(b.III) 35.93%; 2(b.IV) 44.08%; 2(b.V) 49.83%; and 2(b.VI) cumulative trend] compared with *S. Typhimurium* (WT) [Figure 2(b.III)

30.62%; 2(b.IV) 37.58%; 2(b.V) 36.06%; and 2(b.VI) cumulative trend] was observed in the presence of meropenem. The complementation of *ompA* in knockout bacteria efficiently reversed the depolarization phenotype [Figure 2(b.III) 8.26%; 2(b.IV) 8.48%; 2(b.V) 22.07%; and 2(b.VI) cumulative trend]. Collectively, our results suggested that the lack of *OmpA* increases the outer membrane depolarization of bacteria in response to antibiotics. We speculated that the increased antibiotic-mediated depolarization might hamper the survival of the bacteria. To verify our hypothesis, we stained the bacteria with propidium iodide.³⁶ It was observed that in all three concentrations of ceftazidime [Figure 2(c.III), (c.IV), (c.V) and (c.VI)] and meropenem [Figure 2(d.III), (d.IV), (d.V) and (d.VI)], a significantly higher percentage of *S. Typhimurium* $\Delta ompA$ [ceftazidime: Figure 2(c.III) 80.76%; (c.IV) 83.34%; (c.V) 82.31%; and (c.VI) cumulative trend; and meropenem: Figure 2(d.III) 28.31%; (d.IV) 38.55%, (d.V) 42.66%; and (d.VI) cumulative trend] takes up propidium iodide compared with *S. Typhimurium* (WT) [ceftazidime: Figure 2(c.III) 66.72%; (c.IV) 66.99%; (c.V) 76.16%; and (c.VI)

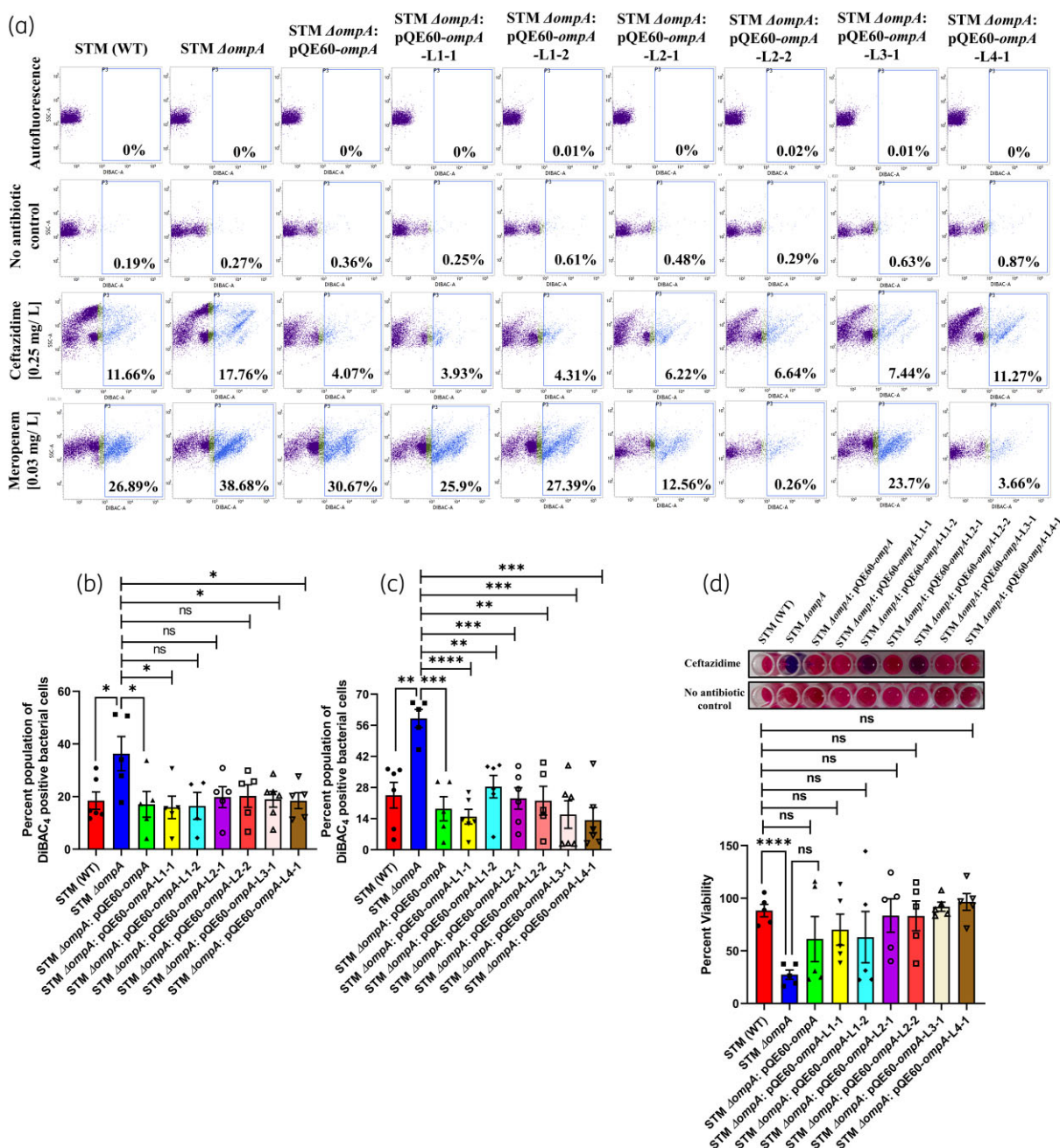


Figure 3. None of the extracellular loops but the complete structure of OmpA protects *S. Typhimurium* from the effect of antibiotics. (a) Estimation of the outer membrane depolarization of *S. Typhimurium* (WT), Δ ompA, Δ ompA:pQE60-ompA, Δ ompA:pQE60-ompA L1-1, Δ ompA:pQE60-ompA L1-2, Δ ompA:pQE60-ompA L2-1, Δ ompA:pQE60-ompA L2-2, Δ ompA:pQE60-ompA L3-1 and Δ ompA:pQE60-ompA L4-1 in the presence of ceftazidime (0.25 mg/L) and meropenem (0.03 mg/L) by flow cytometry ($n=2$, $N=3$). The dot plot (SSC-A versus DCFDA-A) and the histogram (count versus DCFDA-A) have been obtained from BD FACSuite software. Graphical representation of the DIBAC₄-positive population for (b) ceftazidime and (c) meropenem. (d) Measurement of the percent viability of *S. Typhimurium* (WT), Δ ompA, Δ ompA:pQE60-ompA, Δ ompA:pQE60-ompA L1-1, Δ ompA:pQE60-ompA L1-2, Δ ompA:pQE60-ompA L2-1, Δ ompA:pQE60-ompA L2-2, Δ ompA:pQE60-ompA L3-1 and Δ ompA:pQE60-ompA L4-1 upon ceftazidime treatment by resazurin assay ($n=2$, $N=3$). Data are represented as mean $\pm$ SEM. * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; **** $P < 0.0001$; ns, non-significant (unpaired Student's *t*-test). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

cumulative trend; and meropenem: Figure 2(d.III) 3.86%; (d.IV)-25.25%; (d.V) 20.26%; and (d.VI) cumulative trend] and Δ ompA:pQE60-ompA (ceftazidime: Figure 2(c.III) 45.61%; (c.IV) 72.8%; (c.V) 73.91%; and (c.VI) cumulative trend; and meropenem: Figure 2(d.III) 5.22%; (d.IV) 25.75%; (d.V) 32.92%; and (d.VI) cumulative trend), which suggested enhanced killing of

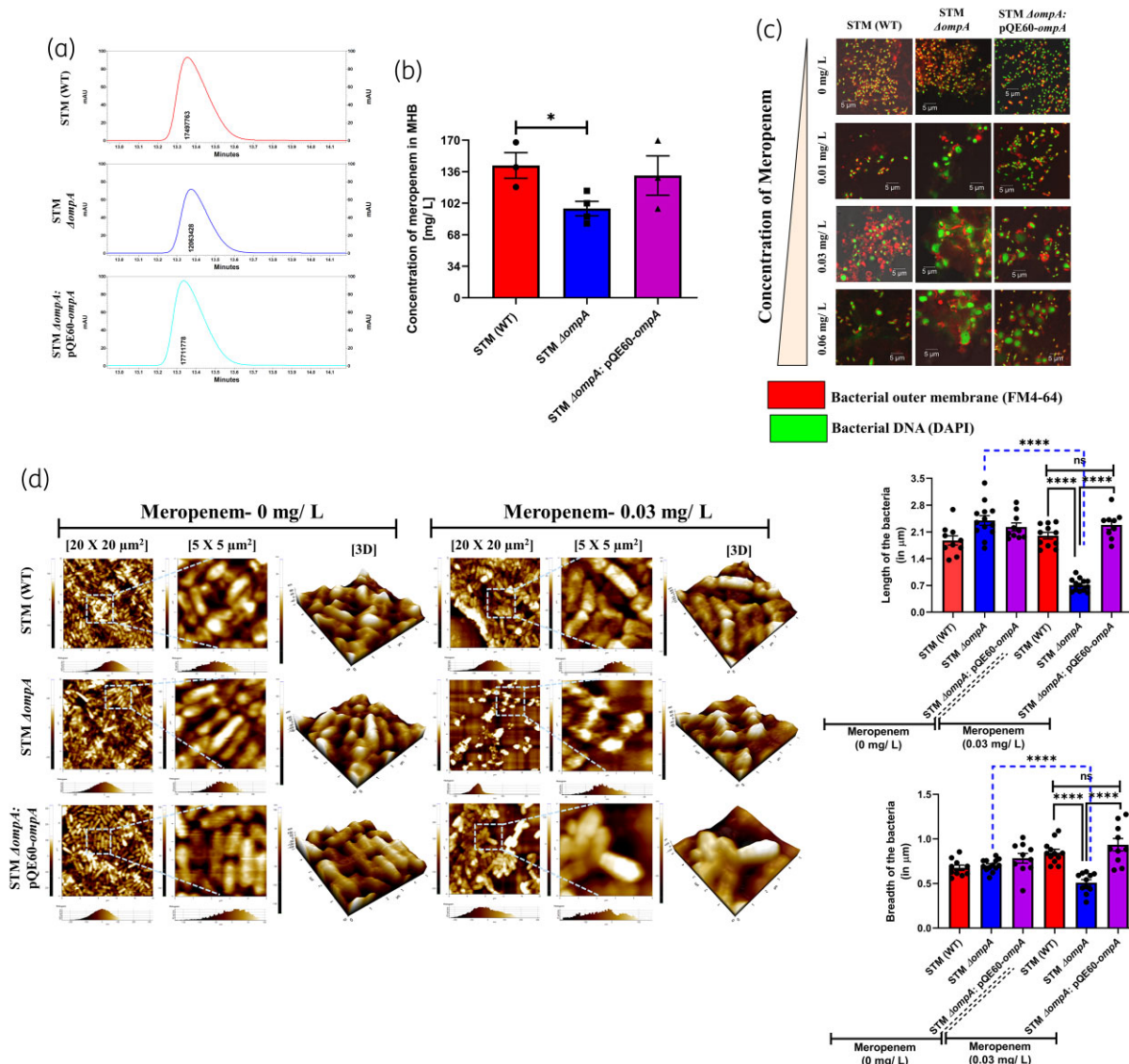


Figure 4. The outer membrane of *S. Typhimurium* $\Delta ompA$ is severely damaged due to increased uptake of antibiotics. (a) Graphical representation and (b) quantification of the amount of meropenem entering the log phase cultures of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA$:pQE60-ompA ($N=4$). (c) Pictorial representation of the outer membrane damage of *S. Typhimurium* (WT), $\Delta ompA$ and $\Delta ompA$:pQE60-ompA, growing in the presence of increasing concentrations of meropenem (0, 0.01, 0.03 and 0.06 mg/L, respectively) ($N=3$). The outer membrane and the genomic DNA of the bacteria were stained with FM 4-64 and DAPI, respectively. The representative image represents one single experiment of three independently done experiments. Scale bar = 5 μm . (d) Atomic force micrograph to study the bacterial morphology in the presence and absence of meropenem. For image acquisition, a $20 \times 20 \mu m^2$ area from each coverslip containing desiccated bacterial samples (antibiotic-treated or untreated) was used. The length and the breadth of the bacteria under meropenem treatment were represented in the form of a bar graph. Data are represented as mean $\pm$ SEM. * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; ns, non-significant (unpaired Student's t -test). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

Salmonella by antibiotic in the absence of OmpA. Bactericidal antibiotics induce oxidative stress to kill pathogens.^{37,38} The ROS-inducing ability of ceftazidime and meropenem has been tested in many pathogens.^{39–41} We hypothesized that, apart from membrane depolarization, the increased burden of intracellular ROS might have an additional role in killing *S. Typhimurium* $\Delta ompA$. The DCFDA staining of antibiotic-treated *S. Typhimurium* (WT) and $\Delta ompA$ suggested the production of an equivalent

amount of ROS [ceftazidime: Figure S3(a.I–a.III); *S. Typhimurium* (WT) 64.33%; *S. Typhimurium* $\Delta ompA$ 64.26%; and meropenem: Figure S3(b.I–S3b.III); *S. Typhimurium* (WT) 12.8%; *S. Typhimurium* $\Delta ompA$ 11.29%]. To correlate the antibiotic-mediated killing of *S. Typhimurium* $\Delta ompA$, we further investigated the expression of oxidative stress response genes of *Salmonella*.^{42–45} We have seen that deleting *ompA* down-regulated the transcription level of *sodA*, *sodB*, *katG* and *katE* in *Salmonella*, which was restored

upon complementation of *ompA* [Figure S3(c)]. We concluded that irrespective of the presence or absence of OmpA, the bacteria experience an equivalent amount of oxidative stress upon antibiotic treatment. However, the higher outer membrane depolarization and the reduced expression of oxidative stress response genes in the absence of *ompA* result in the antibiotic-mediated killing of the bacteria.

None of the extracellular loops but the complete structure of OmpA protects *S. Typhimurium* from the effect of antibiotics

Previously, we reported that introducing mutations in the extracellular loops of OmpA does not alter its folding, expression and outer membrane localization in *Salmonella*.⁴⁶ We hypothesized that these extracellular loops could compensate for the function of complete OmpA in providing antibiotic resistance. We subjected the OmpA extracellular loop mutants (*S. Typhimurium* Δ *ompA*:pQE60-*ompA* L1-1, L1-2, L2-1, L2-2, L3-1 and L4-1) to antibiotic treatment [Figure S4(a and b)]. The better survival of the OmpA loop mutants than *S. Typhimurium* Δ *ompA* in the presence of ceftazidime and meropenem proved that none of these extracellular loops could compensate for the function of complete OmpA protecting the bacteria from antibiotics [Figure S4(a and b)]. Compared with the WT (ceftazidime 11.66%; meropenem 26.89%) and the complemented (ceftazidime 4.07%; meropenem 30.67%) strains, *S. Typhimurium* Δ *ompA* showed enhanced outer membrane depolarization (ceftazidime 17.76%; meropenem 38.68%) upon antibiotic treatment, which was significantly reduced in the OmpA loop mutants (ceftazidime: *S. Typhimurium* Δ *ompA*:pQE60-*ompA* L1-1 3.93%; L1-2 4.31%; L2-1 6.22%; L2-2 6.64%; L3-1 7.44%; and L4-1 11.27%; and meropenem: L1-1 25.9%; L1-2 27.39%; L2-1 12.56%; L2-2 0.26%; L3-1 23.7%; and L4-1 3.66%) [Figure 3(a, b and c)]. Under ceftazidime treatment, the increased percent viability of the OmpA loop-mutants compared with *S. Typhimurium* Δ *ompA* strongly corroborated this observation [Figure 3(d)]. Hence, we concluded that instead of the individual extracellular loops, the complete membrane-bound structure of OmpA is required for the antimicrobial protection of *Salmonella*.

Outer membrane of *S. Typhimurium* Δ *ompA* is severely damaged due to increased uptake of antibiotics

We speculated that in the absence of OmpA, β -lactam antibiotics would penetrate more into the bacteria and cause severe damage to the outer membrane. To test our hypothesis, *S. Typhimurium* (WT), Δ *ompA* and Δ *ompA*:pQE60-*ompA* were treated with very high concentrations of meropenem (100–150 mg/L) and ceftazidime (300 mg/L) for 1 h. The remaining concentration of antibiotics was quantified by HPLC to estimate the antibiotic consumption by the bacteria [Figures 4(a and b), S5(a and b), S6(a and b)]. Our study revealed a higher intake of meropenem and ceftazidime by *S. Typhimurium* Δ *ompA* compared with *S. Typhimurium* (WT) [Figure 4(a and b), S5(a and b), S6(a and b)]. To show the membrane disruption of *Salmonella* upon antibiotic consumption, meropenem-treated bacteria were stained with DAPI and FM 4-64 [Figure 4(c)]. The genomic DNA (green) of *Salmonella* was tightly packed inside the intact

membrane without antibiotics [Figure 4(c)]. With an increase in the concentration of meropenem, massive damage to the outer membrane of *S. Typhimurium* Δ *ompA* was observed [Figure 4(c)], which ultimately released the DNA and killed the bacteria. We further assessed the morphology of the bacteria treated with meropenem (0.03 mg/L) by AFM [Figure 4(d)]. The significant reduction in the length and breadth of *S. Typhimurium* Δ *ompA* suggested impaired bacterial morphology under meropenem treatment [Figure 4(d)]. The coordinated expression of several efflux pump genes (*acrB*, *acrF*, *acrD* etc.) and a stress-response transcriptional activator *marA* give rise to multiple antibiotic resistance phenotypes in *Salmonella* Typhimurium SL1344 clinical isolates.⁴⁷ As *S. Typhimurium* Δ *ompA* is consuming more antibiotics, we speculate that the reduction in its viability is associated with its inability to synthesize the efflux pumps. We observed that *S. Typhimurium* Δ *ompA* is deficient in producing *acrF* and *marA* compared with the WT and the complemented bacteria [Figure S7(a)]. We finally concluded that *Salmonella* lacking OmpA consumes more antibiotics and experiences significant damage to its outer membrane. Additionally, the compromised expression of *marA* and *acrF* enhanced the lethality of the antibiotics on *S. Typhimurium* Δ *ompA*.

Administration of ceftazidime cleared bacterial infection from C57BL/6 mice more efficiently than meropenem

Besides antimicrobial resistance, bacterial cells have developed alternative strategies to survive antibiotic stress. Prolonged treatment of antibiotics in the infected hosts can generate antibiotic persisters, which constitute the transiently drug-tolerant non-proliferating phenotypic variants within isogenic populations.^{48–51} Membrane depolarization can also induce persistence in bacteria.⁵² However, the impact of outer membrane depolarization on the persistence of *S. Typhimurium* is yet to be investigated. We hypothesized that the increased membrane depolarization of *S. Typhimurium* Δ *ompA* would generate a more significant fraction of antibiotic persisters. Surprisingly, we found that deleting *ompA* resulted in a lesser percent viability of persisters compared with WT in both planktonic [Figure 5(a)] as well as in biofilm [Figure 5(c)] culture under prolonged ceftazidime (50 $\times$ MIC) treatment. However, no significant difference was reflected in the percent viability of *S. Typhimurium* (WT) and *S. Typhimurium* Δ *ompA* in both planktonic [Figure 5(b)] and biofilm [Figure 5(d)] cultures following exposure to meropenem (25 $\times$ MIC). We subjected 4–6-week-old C57BL/6 mice [infected with 10⁶ cfu of *S. Typhimurium* (WT) and Δ *ompA*] to antibiotic treatment [Figure 5(e–g)]. Ceftazidime and meropenem (5 mg/kg of body weight) were administered to the infected mice on the second and fourth day post-infection by intraperitoneal injection [Figure 5(e)]. On the fifth day, the mice were sacrificed to enumerate the bacterial load in organs. It was found that, unlike meropenem, the administration of ceftazidime efficiently reduced the burden of *S. Typhimurium* Δ *ompA* in the liver and spleen of C57BL/6 mice compared with the antibiotic-untreated mice [Figure 5(f and g)].

Discussion

The rapid emergence of the drug resistance phenotype in nontyphoidal serovars of *Salmonella* is a significant obstacle in curing

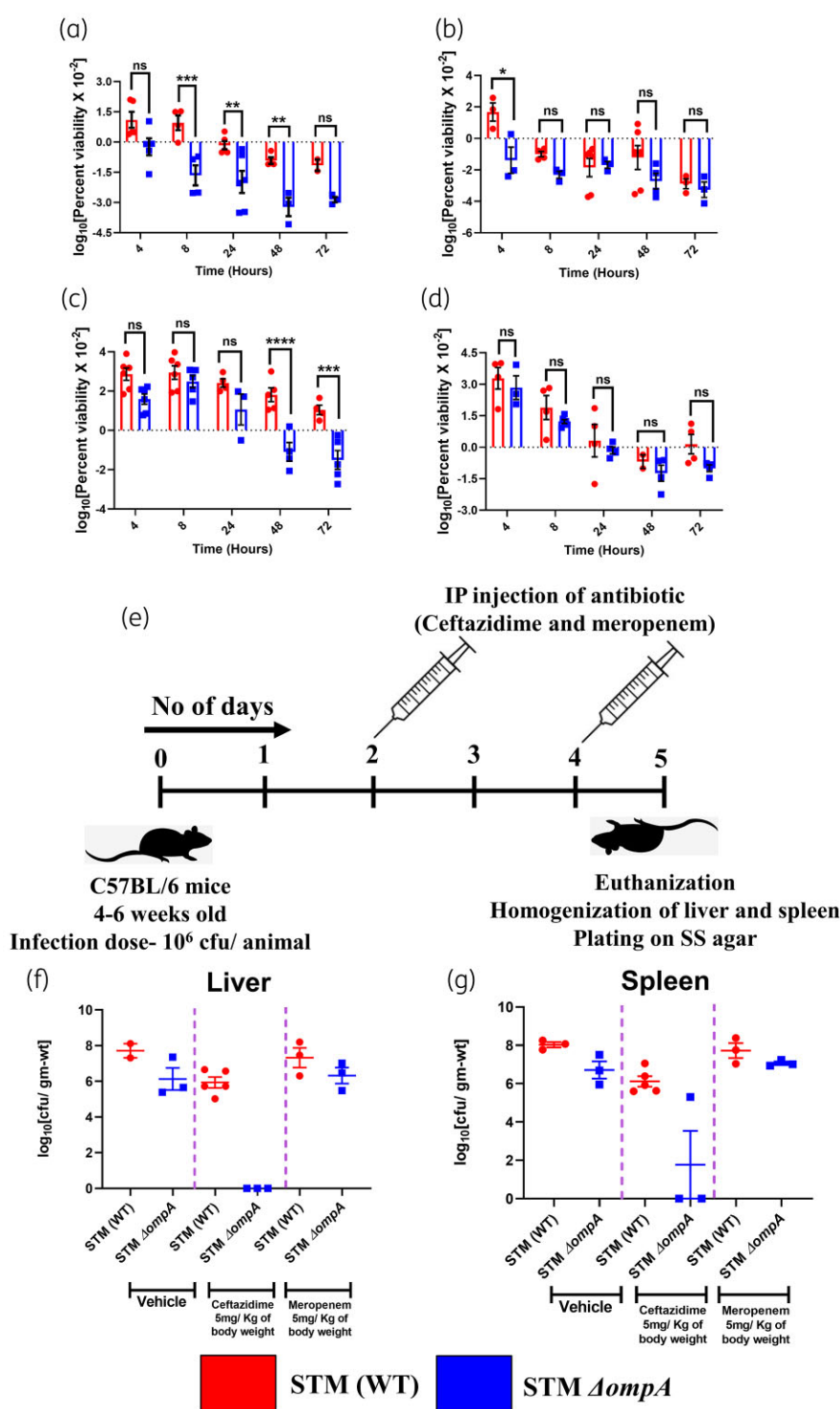


Figure 5. Organ burden of *S. Typhimurium* (WT) and $\Delta ompA$ under the treatment of ceftazidime and meropenem (5 mg/kg of body weight). (a–d) Calculating the antibiotic-tolerant persister fraction of *S. Typhimurium* (WT) and $\Delta ompA$ growing in planktonic culture (a, b) and biofilm (c, d) in the presence of ceftazidime (50 mg/L; 50 $\times$ MIC) (a and c) and meropenem (50 mg/L 25 $\times$ MIC) (b and d) for 72 h ($n=2$, $N=3$). (e–g) 4–6-week-old C57BL/6 mice were infected with 10^6 cfu of *S. Typhimurium* (WT) and *S. Typhimurium* $\Delta ompA$ ($n=5$). The mice were treated with ceftazidime and meropenem (5 mg/kg of body weight) on the specified days (e). On the fifth day post-infection, the mice were sacrificed. The liver (f) and spleen (g) were collected and homogenized. The cell lysates were plated to enumerate the bacteria load in each organ. The cfu obtained from the liver and spleen were normalized with the weight of the individual organs. Data are represented as mean $\pm$ SEM. (P) * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; **** $P < 0.0001$; ns, non-significant (two-way ANOVA). This figure appears in colour in the online version of JAC and in black and white in the print version of JAC.

Salmonella-induced foodborne illnesses with antibiotic therapy.^{53,54} Tigecycline and carbapenems are the latest drugs to treat MDR and XDR typhoid fever. However, the continuous adaptation of the pathogen creates a potential risk of developing resistance against recommended antibiotics in the future, providing an opportunity to study new drugs and their potential targets in detail. Many Gram-negative pathogens use outer membrane porins to maintain outer membrane stability and develop antibiotic resistance by regulating permeability.^{12,15,16,18,55–60} In the current study, we have tried to delineate the contributions of the four most abundant outer membrane porins of *S. Typhimurium*, such as OmpA, OmpC, OmpD and OmpF, in developing resistance against two broad-spectrum β -lactam antibiotics, ceftazidime and meropenem.^{26,61–64} The reduction in the MIC of both antibiotics for *S. Typhimurium* Δ ompA suggested that *S. Typhimurium* OmpA plays an essential role in protecting the pathogen from the β -lactam antibiotics. Unlike other porins such as OmpC, OmpD and OmpF, apart from having an outer membrane-bound β -barrel subunit, *S. Typhimurium* OmpA has a periplasmic domain that interacts with the peptidoglycan layer, thus maintaining the stability of the outer membrane in the presence of antibiotics.^{13,65} Similar studies showing the OmpA-dependent protection of *Acinetobacter baumannii* from β -lactam antibiotics corroborated our findings.^{66,67}

The enhanced outer membrane depolarization and the killing of *S. Typhimurium* Δ ompA afterward proved the pivotal role of OmpA in protecting *Salmonella* from the β -lactam antibiotics by maintaining the stability of its outer membrane. Bacteria use oxidative stress response genes for protection against antibiotic-induced oxidative damage.^{68–70} In addition to higher membrane depolarization, the reduced expression of *sod* (*sodA*, *sodB*) and *kat* (*katG*, *katE*) genes in *S. Typhimurium* Δ ompA explained their incapability to tolerate antibiotic-driven oxidative stress, unlike *S. Typhimurium* (WT). In continuation with this, we found that *S. Typhimurium* Δ ompA consumes more antibiotics than WT bacteria. *A. baumannii* OmpA directly interacts with several efflux pumps to drive out antibiotics from the periplasm.^{71,72} The reduction in the transcript level of *marA* (a stress-induced transcriptional activator) and *acrF* (efflux pump) in *S. Typhimurium* Δ ompA indicates the inability of the mutant bacteria to force out antibiotics from the periplasm and receive more damage than WT *Salmonella*. Previous studies in *Escherichia coli* and *Staphylococcus aureus* demonstrated an association between membrane depolarization and the generation of antibiotic persisters. Much in disparity, we found that deleting *ompA* from *Salmonella* decreased persister cell levels until 72 h post-exposure to ceftazidime (50 mg/L). The enhanced killing of *S. Typhimurium* Δ ompA upon ceftazidime treatment might override the antibiotic persistence phenotype. The better clearance of *S. Typhimurium* Δ ompA by ceftazidime during *in vivo* infection proved the protective role of *S. Typhimurium* OmpA against β -lactam drugs.

Conclusions

To summarize, our study demonstrates that deleting OmpA weakens the outer membrane of *Salmonella* and facilitates the entry of two β -lactam antibiotics (ceftazidime and meropenem),

which ultimately kill the bacteria. Further, we showed that *S. Typhimurium* Δ ompA is deficient in expressing the oxidative stress-response and efflux pump genes, which makes the bacteria highly susceptible to antibiotic-mediated damage.

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Transparency declarations

None to declare.

Author contributions

A.R.C. and D.M. equally contributed to the manuscript. A.R.C. and D.C. conceived the study and designed the experiments. A.R.C. and D.M. performed all the experiments and participated in acquiring the data. A.S. performed the experiments with A.R.C. and D.M. A.R.C. analysed the data, constructed the figures, and wrote the original draft of the manuscript. D.M. participated in the proofreading and editing of the manuscript. D.C. supervised the study. All the authors have read and approved the manuscript.

Supplementary data

Supplementary data at JAC Online.

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